

LED Pulse Generator

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Main parameters

Form factor	NIM unit
Number of Channels	30
Mode of operation	external/internal generator
Internal Generator Frequency	5kHz-7kHz
External Generator Input	positive TTL Level, impedance 50 Ohm
Outputs	positive TTL Level, 42mA (50 Ohm load)
Output Pulse Width (from internal generator)	7ns-400ns
Front end input-output connectors	Lemo type
DC Power Supply	+6V at 450mA

Description

The LED Pulse Generator is developed for testing of scintillator detectors with LEDs as pulsed source of the light. The amount of the light emitted by the LED is defined by the width of the output pulses from generator.

Generator consists of input logic, output drivers and short pulses internal generator with regulated pulse width. The mode of operation is selected by switch located on the front panel. The width of the output pulses from internal generator is regulated manually by potentiometer located also on the front panel.

The input connector "Inp" for external generator and fun-out connector "Out" are located at the top of the module. In internal generator mode the input works as enable/disable input for internal generator. The output drivers based on Philips 74F3037N chips.

Operation

From External Generator

Turn mode operation switch in "Ext." position. Connect output with TTL level from external generator to the input connector of the module("Inp"). The output pulse rate and pulse width from the module will be determined by the pulses from the external generator.

From Internal Generator

Turn mode operation switch in "Int." position. To enable the internal generator send to the input TTL signal with high level (logic 1). To disable the internal generator send to the input TTL signal with low level (logic 0). The amount of the light emitted by LED is defined by the setting of potentiometer on front panel.

To turn on the internal generator continuously in the case if there is no any signal on the input and no connection to the input connector some one should set the jumper J1 located inside of the module to the open position. In this position of the jumper the input impedance of the module is 10 kOhm. To turn off the internal generator, disconnect from the input any signals and set the jumper J1 located inside of the module to the close position. In this position of the jumper the input impedance of the module is 50 Ohm.

Pic.1. Front Panel of the LED Pulse Generator

